DESIGN AND APPLY CONTEXT-AWARE ACCESS CONTROLS BASED ON USER IDENTITY, DEVICE POSTURE, LOCATION AND APPLICATION RISK

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INTRODUCTION:

With the rapid growth of internet technologies and online services, securing web applications has become a critical concern for organizations. Many systems still rely on traditional authentication methods such as usernames and passwords to control access to resources. Although these methods are widely used, they are not always sufficient to protect systems from security threats.

Cyber attackers often exploit vulnerabilities in authentication systems to gain unauthorized access. For example, if login credentials are stolen or guessed, attackers can easily access sensitive data. Therefore, relying solely on user credentials is no longer considered a secure solution for modern applications.

Context-aware access control is an advanced security approach that considers additional contextual information before allowing access. Instead of simply verifying the identity of the user, the system evaluates multiple factors such as the user's device, network location, and the risk level of the application being accessed.

By analyzing these contextual parameters, the system can make more intelligent security decisions. For example, access may be allowed if the user is using a secure device within the office network, but it may be restricted or denied if the same user tries to access the system from an external network using an unsecure device.

The Context-Aware Access Control System developed in this project demonstrates how contextual security policies can be implemented in a web application environment. The system provides a user-friendly interface and evaluates different contextual factors to determine the appropriate access level.

ABSTRACT

In modern web applications, protecting sensitive data and preventing unauthorized access have become major security challenges. Traditional access control mechanisms mainly rely on username and password authentication, which may not be sufficient to ensure secure access. Attackers may gain unauthorized access if login credentials are compromised. To overcome this limitation, context-aware access control systems are introduced.

The Context-Aware Access Control System enhances security by evaluating multiple contextual parameters before granting access to a user. These parameters include user identity, device security posture, network location, and application risk level. Instead of making access decisions based solely on authentication credentials, the system dynamically evaluates the user's environment and security conditions.

The proposed system is implemented using HTML, CSS, and JavaScript to create an interactive web interface. The system collects contextual information from users and processes it using predefined security policies to determine whether access should be granted, restricted, or denied.

The system is tested using the vulnerable web application testphp.vulnweb.com to simulate real-world access scenarios. The project demonstrates how context-based decision making can significantly improve the security of web applications and reduce the risk of unauthorized access.

SCENARIO OF THE SYSTEM

The Context-Aware Access Control System evaluates different contextual parameters before granting access to a user. The following scenarios demonstrate how the system behaves under different conditions.

Scenario 1: Secure Access from Trusted Environment

- In this scenario, a user attempts to access the system from a secure device within the office network. The user role is selected as admin and the application risk level is low.
- Since the user is accessing the system from a trusted environment and the device meets the security requirements, the system grants access successfully. The access decision is recorded in the access history table along with the username and timestamp.
- This scenario represents a normal and secure access situation where all contextual conditions meet the security policy.

Scenario 2: Access from External Network

- In this scenario, a user attempts to access the system from an external network using a secure device. The application risk level is set to high.

- Because the user is accessing the system from outside the trusted network, the system identifies this as a potential risk. Depending on the security policy, the system may provide restricted access or require additional verification.
- The system records the attempt in the access log and displays the access status accordingly.

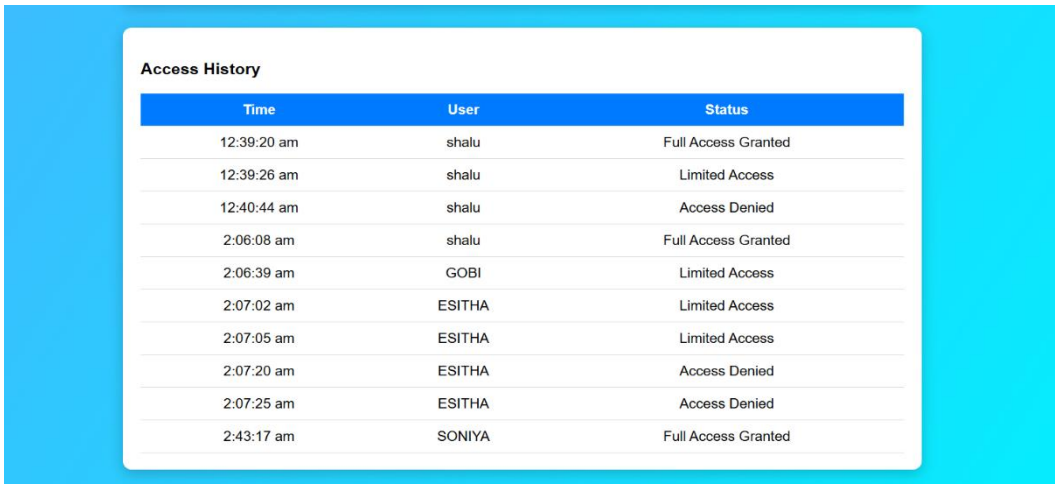
Scenario 3: Unsecure Device Access Attempt

- In this scenario, a user attempts to access the system using an unsecure device. The device may lack proper security protection or may be considered vulnerable.
- Even if the user has valid credentials, the system identifies the device posture as a security risk. Therefore, the system denies access in order to protect sensitive data and prevent potential attacks.
- The denied access attempt is recorded in the log table for monitoring and auditing purposes.

Scenario 4: High Risk Application Access

- In this scenario, a user attempts to access a high-risk application from an external network using an unsecure device.
- Since multiple risk factors are present, the system classifies this request as a high-risk access attempt. As a result, the system denies access to prevent unauthorized usage.
- This scenario demonstrates how the context-aware system makes intelligent decisions by evaluating multiple parameters simultaneously.

Context Aware Access Control



Access History

Time	User	Status
12:39:20 am	shalu	Full Access Granted
12:39:26 am	shalu	Limited Access
12:40:44 am	shalu	Access Denied
2:06:08 am	shalu	Full Access Granted
2:06:39 am	GOBI	Limited Access
2:07:02 am	ESITHA	Limited Access
2:07:05 am	ESITHA	Limited Access
2:07:20 am	ESITHA	Access Denied
2:07:25 am	ESITHA	Access Denied
2:43:17 am	SONIYA	Full Access Granted

ADVANTAGES OF THE SYSTEM

The Context-Aware Access Control System provides several advantages compared to traditional access control methods. One of the main advantages is enhanced security. By evaluating multiple contextual parameters such as device security, location, and application risk, the system reduces the chances of unauthorized access.

Another advantage is dynamic decision making. The system does not rely on fixed rules based only on user credentials. Instead, it evaluates the user's current context and environment before granting access.

The system also improves monitoring and auditing capabilities by maintaining a record of all access attempts. This allows administrators to analyze user behaviour and identify suspicious activities. Additionally, the system is easy to implement and use because it is developed using simple web technologies such as HTML, CSS, and JavaScript. This makes the system suitable for educational and demonstration purposes.

CONCLUSION

The Context-Aware Access Control System developed in this project demonstrates an effective approach to improving security in web applications. Unlike traditional authentication methods that rely only on usernames and passwords, the proposed system evaluates additional contextual parameters before granting access.

By analyzing factors such as user identity, device posture, network location, and application risk, the system can make intelligent access decisions that enhance security and reduce vulnerabilities.

The implementation of the system using HTML, CSS, and JavaScript provides a simple and interactive environment for demonstrating context-aware security policies. The access logging feature further improves system monitoring by keeping track of all access attempts.

Overall, this project highlights the importance of context-based security mechanisms in modern web applications and shows how they can help protect systems from unauthorized access.

FUTURE ENHANCEMENT

1. Database Integration

In the future, the system can be connected to a database such as MySQL to store user information, login details, and access logs permanently. This will allow better management of multiple users and historical records.

2. Multi-Factor Authentication (MFA)

The system can be improved by adding additional authentication methods such as OTP verification or biometric authentication. This will provide an extra layer of security beyond username and password.

3. Real-Time Device Security Detection

Future versions of the system can automatically check device security status such as antivirus availability, operating system updates, and firewall protection before granting access.

4. Location Tracking Using GPS

Instead of manually selecting the location, the system can automatically detect the user's location using GPS or IP address to make more accurate access decisions.

5. Machine Learning Based Risk Detection

Machine learning algorithms can be integrated to analyze user behavior patterns and detect unusual access attempts automatically.

6. Integration with Enterprise Security Systems

The system can be connected with enterprise identity management systems such as Active Directory or Single Sign-On platforms to support large organizations.

7. Mobile Application Support

A mobile version of the system can be developed so that administrators can monitor access logs and security alerts through smartphones.

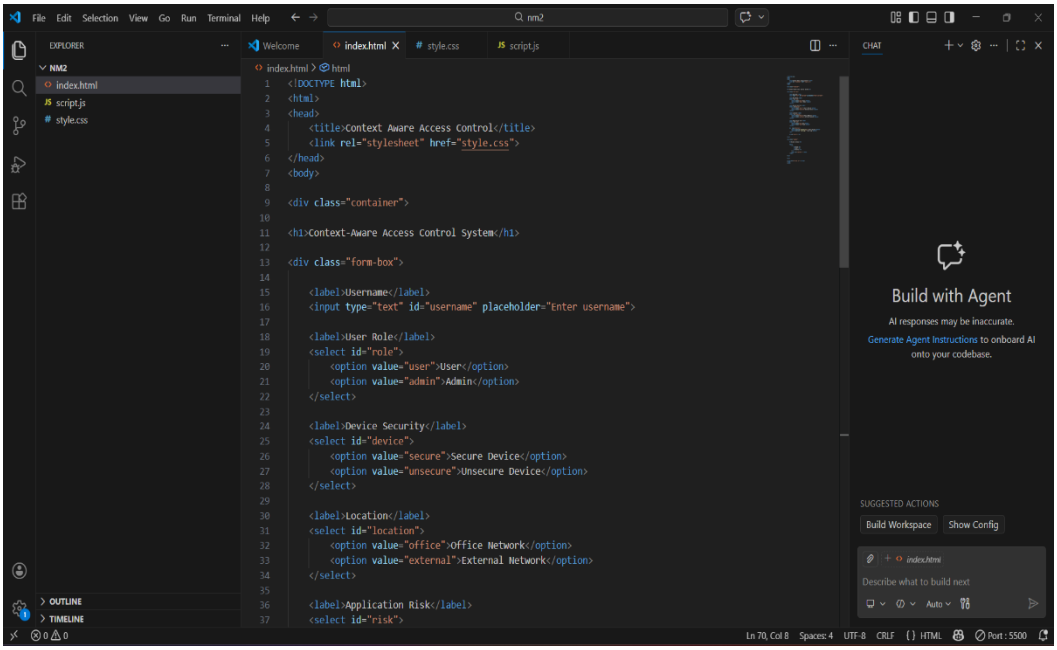
8. Advanced Security Alerts

The system can send automatic alerts through email or SMS when suspicious login attempts are detected.

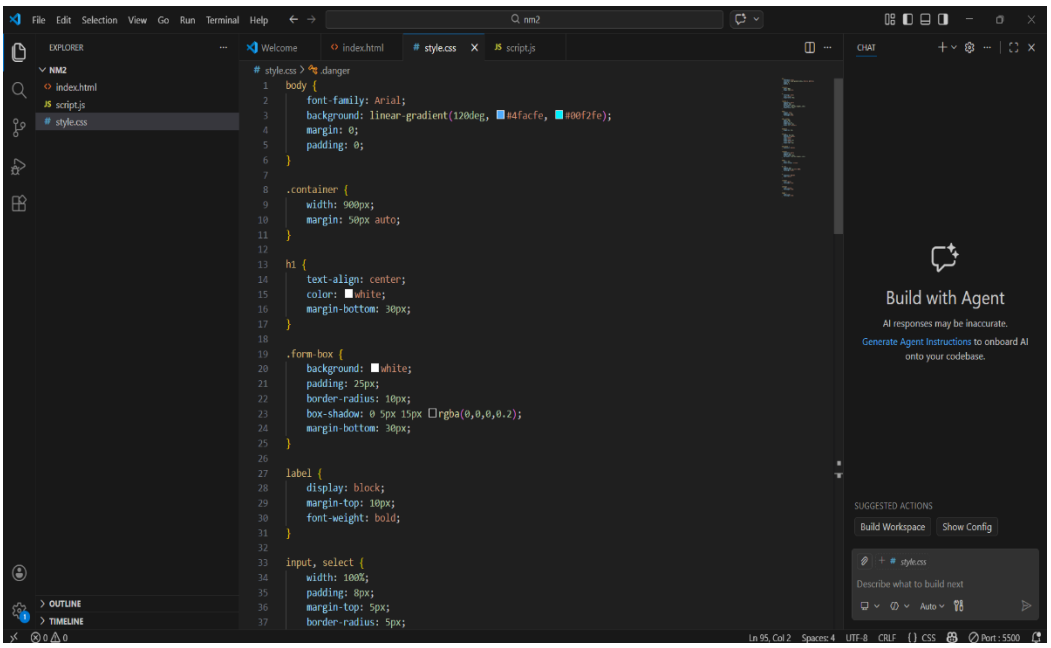
INPUT

USING VISUAL CODE

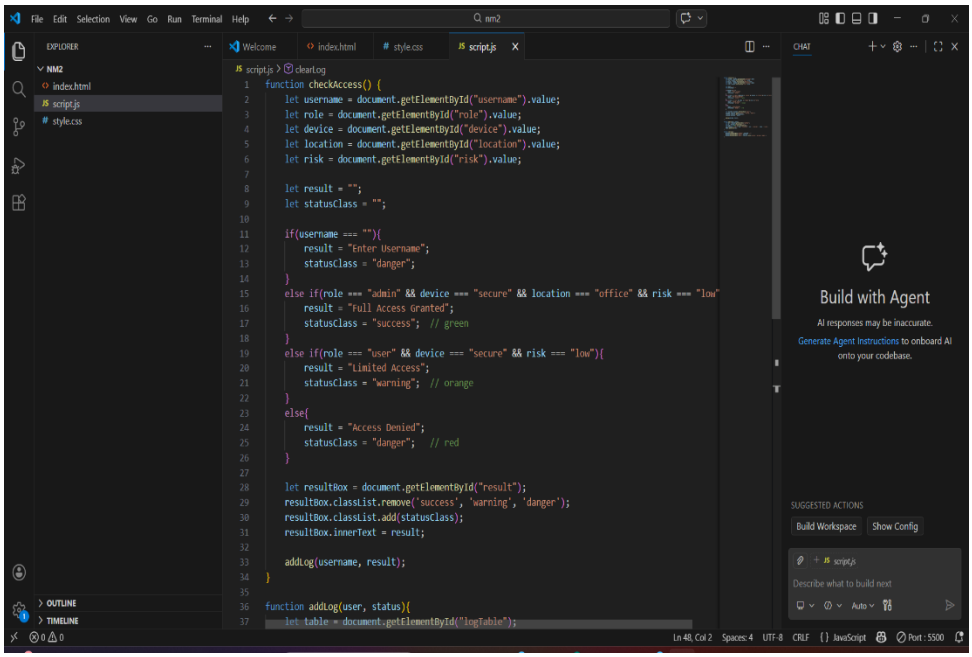
FILE 1: index.html



FILE 2: style.css



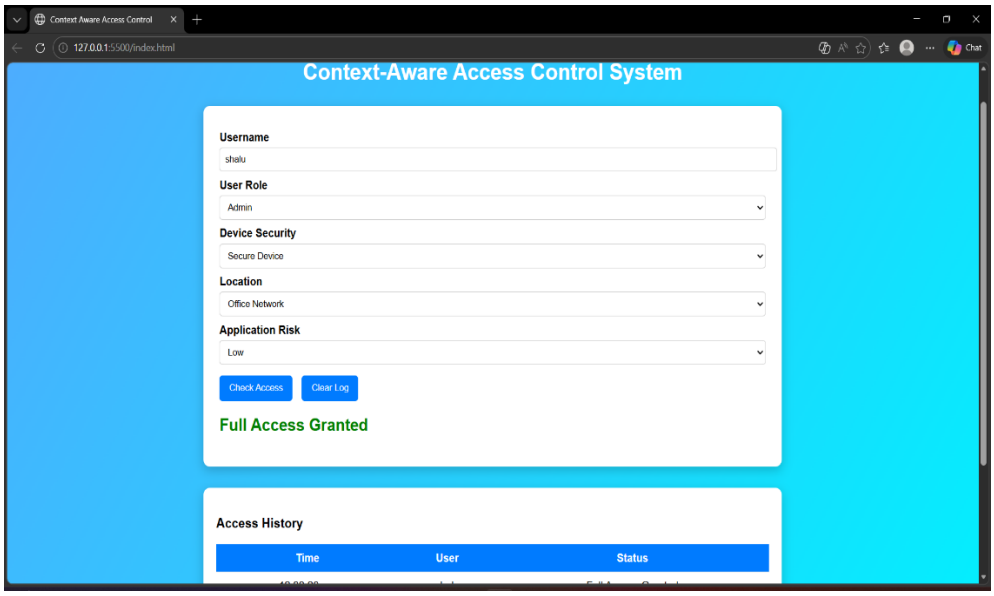
FILE 3: script.js



OUTPUT:

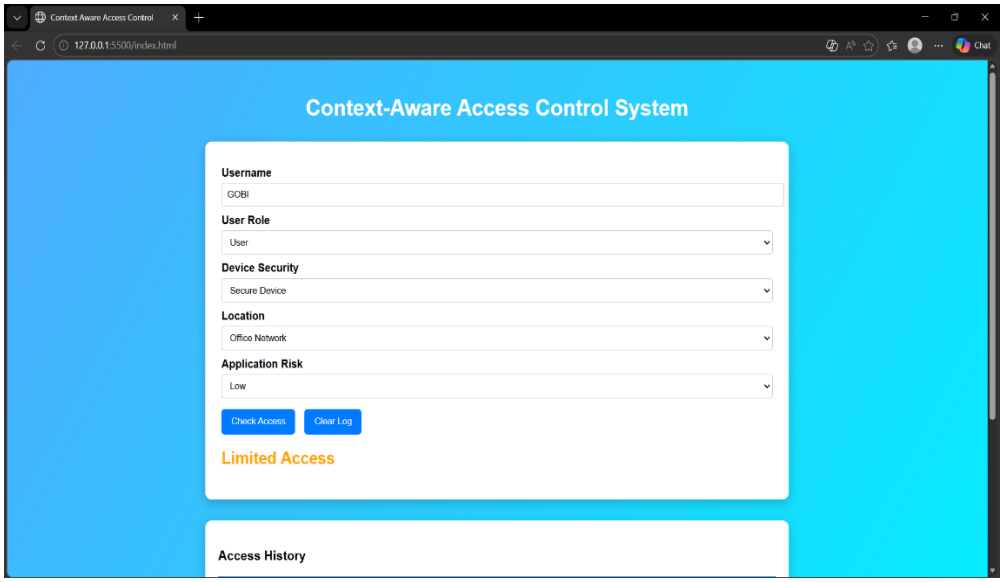
SCREENSHOT 1 :

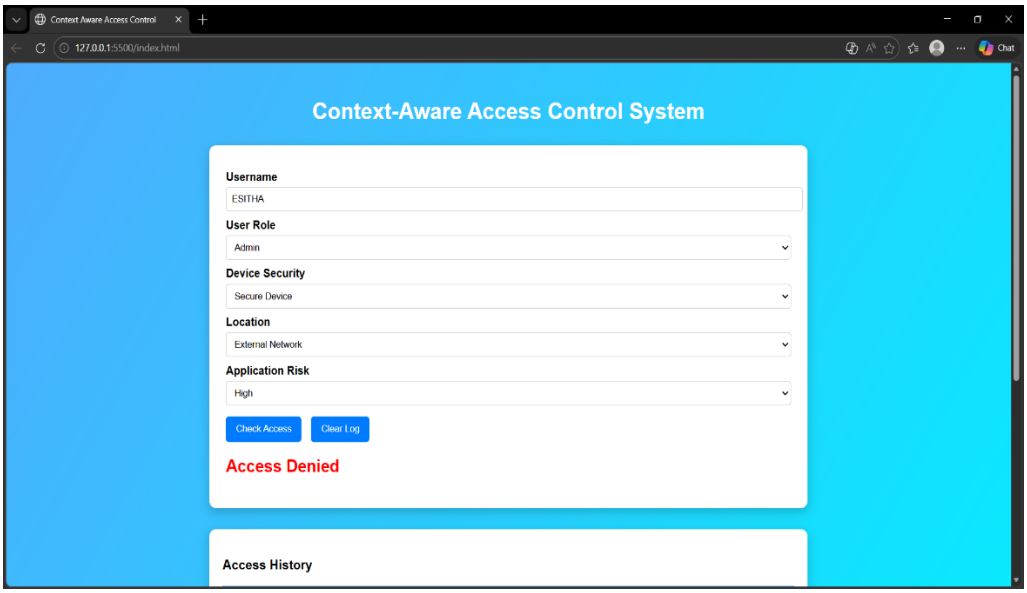
FULL ACCESS GRANTED



SCREEN SHOT 2:

LIMITED ACCESS:



SCREEN SHOT 3:

ACCESS DENIED